MAPS OF COMPLETIONS LIFT ETALE LOCALLY THROUGH CONGRUENCES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof justifies the completion identifications for etale maps, reduces to producing a map into the henselization via the filtered-colimit description, transcribes psi into equations over the completion, and applies the smoothing lemma over the G-ring C to solve them in C^h modulo I^N

Editorial note. The label lemma-fully-faithfulness names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- etale ring maps are flat and induce isomorphisms C/I^nC -> C'/I^nC' for all n (Algebra, Lemma algebra-lemma-etale; More on Algebra, Lemma more-algebra-lemma-neighbourhood-isomorphism)
- the henselization C^h of the pair (C, IC) as the filtered colimit of etale neighbourhoods C', with C -> C' -> C^h inducing isomorphisms on completions (More on Algebra, Lemmas more-algebra-lemma-henselization and more-algebra-lemma-henselization-Noetherian-pair)
- finite type algebras over a Noetherian G-ring are G-rings (More on Algebra, Proposition more-algebra-proposition-finite-type-over-G-ring)
- the Artin approximation statement of Smoothing Ring Maps (Lemma smoothing-lemma-henselian-pair): solutions modulo I^N in the completion lift to solutions in the henselization

Source reference. Original lemma 00U2: sources/stacks-chow/tags.txt, source lines 1096–1096. Prerequisite pointer invoked at proof line(s) 1556; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0A02: sources/stacks-chow/tags.txt, source lines 12620–12620. Prerequisite pointer invoked at proof line(s) 1561; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0AGV: sources/stacks-chow/tags.txt, source lines 13208–13208. Prerequisite pointer invoked at proof line(s) 1565; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

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Source reference. Original lemma 05E9: sources/stacks-chow/tags.txt, source lines 6982–6982. Prerequisite pointer invoked at proof line(s) 1558; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original proposition 07PV: sources/stacks-chow/tags.txt, source lines 9810–9810. Prerequisite pointer invoked at proof line(s) 1574; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0AH5: sources/stacks-chow/tags.txt, source lines 13218–13218. Prerequisite pointer invoked at proof line(s) 1576; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let A be a Noetherian G-ring. Let $I \subset A$ be an ideal. Let B, C be finite type A-algebras. For any A-algebra map $\varphi : B^\wedge \to C^\wedge$ of I-adic completions and any $N \geq 1$ there exist*

(1) *an étale ring map $C \to C'$ which induces an isomorphism $C/IC \to C'/IC'$,*
(2) *an A-algebra map $\varphi : B \to C'$*

such that φ and ψ agree modulo I^N into $C^\wedge = (C')^\wedge$.

Proof. The statement of the lemma makes sense as $C \to C'$ is flat (Algebra, Lemma 00U2) hence induces an isomorphism $C/I^nC \to C'/I^nC'$ for all n (More on Algebra, Lemma 05E9) and hence an isomorphism on completions. Let C^h be the henselization of the pair (C, IC), see More on Algebra, Lemma 0A02. Then C^h is the filtered colimit of the algebras C' and the maps $C \to C' \to C^h$ induce isomorphism on completions (More on Algebra, Lemma 0AGV). Thus it suffices to prove there exists an A-algebra map $B \to C^h$ which is congruent to ψ modulo I^N. Write $B = A[x_1, \ldots, x_n]/(f_1, \ldots, f_m)$. The ring map ψ corresponds to elements $\hat{c}_1, \ldots, \hat{c}_n \in C^\wedge$ with $f_j(\hat{c}_1, \ldots, \hat{c}_n) = 0$ for $j = 1, \ldots, m$. Namely, as A is a Noetherian G-ring, so is C, see More on Algebra, Proposition 07PV. Thus Smoothing Ring Maps, Lemma 0AH5 applies to give elements $c_1, \ldots, c_n \in C^h$ such that $f_j(c_1, \ldots, c_n) = 0$ for $j = 1, \ldots, m$ and such that $\hat{c}_i - c_i \in I^N C^h$. This determines the map $B \to C^h$ as desired. □